

Dictionaries & AI Tools

TIP101

Agenda

1 Welcome 🙌

2 Dictionaries & AI Tools

3 **BREAK**

4 Collaborative Problem Solving

5 Wrap Up

10 minutes

35 minutes

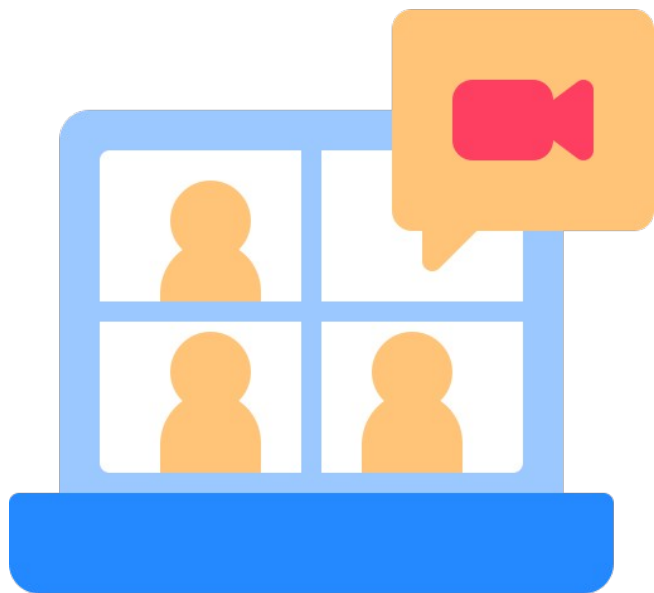
5 minutes

60 minutes

10 minutes

10 minutes

Welcome! 🖐️



**Turn on your
cameras**



**Fill out the entry
ticket**



Announcements & Reminders

* Assessment Reminder:

- Students **must submit one** attempted HackerRank assessment **weekly** to remain in the course (due Sunday at 11:59pm PST).
- If a student moves ahead early in the units, and finishes passes all 10 units, they no longer require weekly attempts, but attendance is still mandatory.
- Keep on eye on the Zoom chat during breakout rooms.

Your Feedback



Glow

- + Breakout rooms going well!
- + Students are finding cheat sheets and other resources helpful.



Grow

- Need more prep material.
- Some students asking for more data structure explanations.



Icebreaker

5 minutes

Question:

Let's settle it once and for all. What truly is the best ice cream flavor? (And what's the worst one 😬?)

Answer:

Share your answer in the Zoom chat below. Multiple options are encouraged.

Choosing Your Breakout Room

1. We'll share the link to the entry ticket so you can indicate what unit you are on.
2. We'll create breakout groups based on the unit you selected.
3. When it's Collaborative Problem Solving time, we'll share the room assignments and demonstrate how to get to your room.



35 minutes

Dictionaries & AI Tools



Check for Understanding

5 minutes

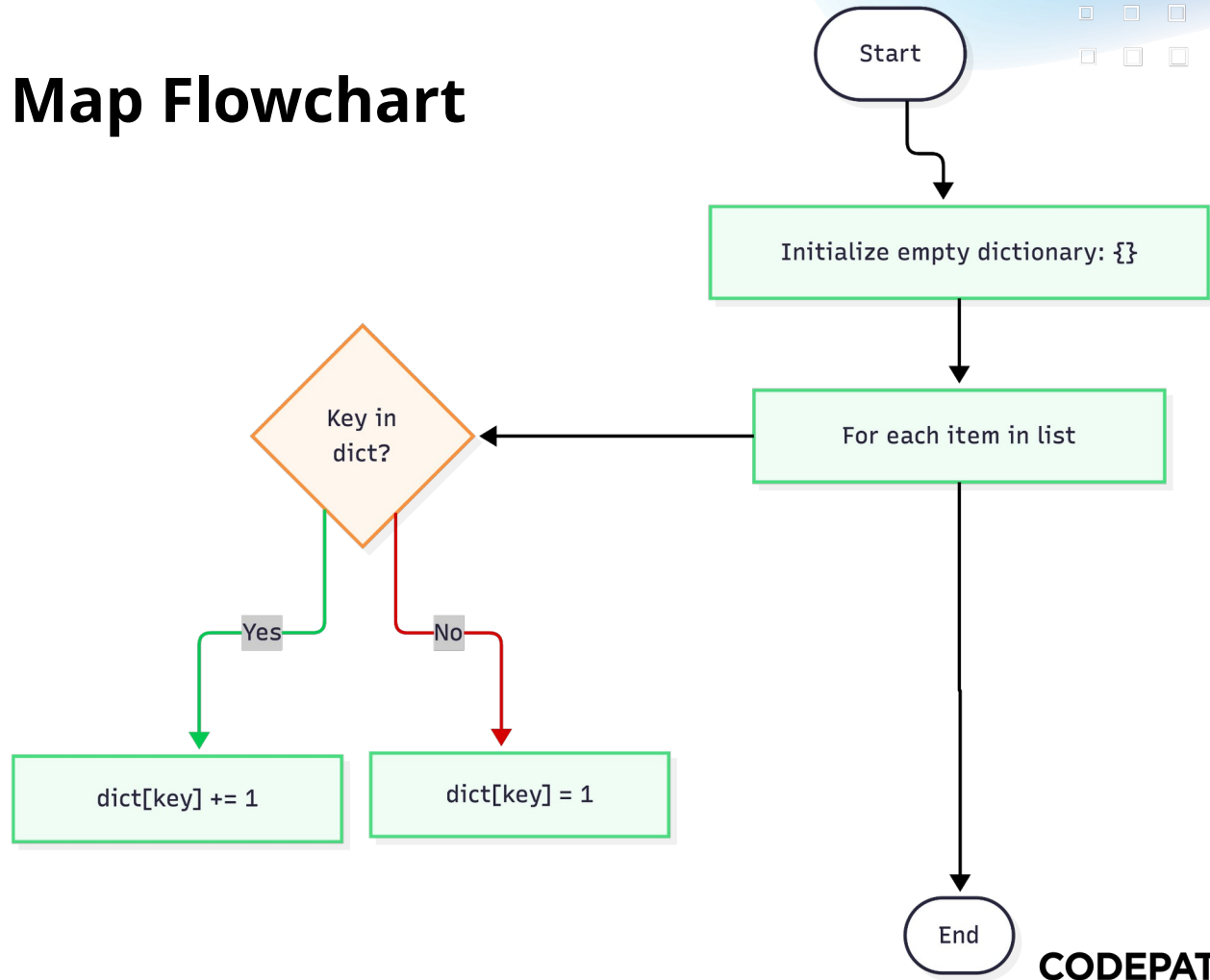
When is it best to use a list?

What about a dictionary?

Understand When to Use Each

- * Lists are great for **ordered** data and iteration:
 - Keeping data in chronological order.
 - Arrival time matters (first come first served).
 - Size expected to be constant (12 slots for monthly revenue).
- * Dictionaries for key-value **mapping**
 - Dynamically grows as new data is added.
 - Speed is prioritized.
 - Unknown keys (user input).

Frequency Map Flowchart





Discussion

3 minutes

In the chat . . .

- Have you ever used an AI tool like Claude or Copilot for learning?
- How did it help you?
- Any specific classes/project types where it performed best?



Instructor Demo

You are given a list of tuples, where each tuple contains a category and a value. Write a function to count how many items belong to each category.

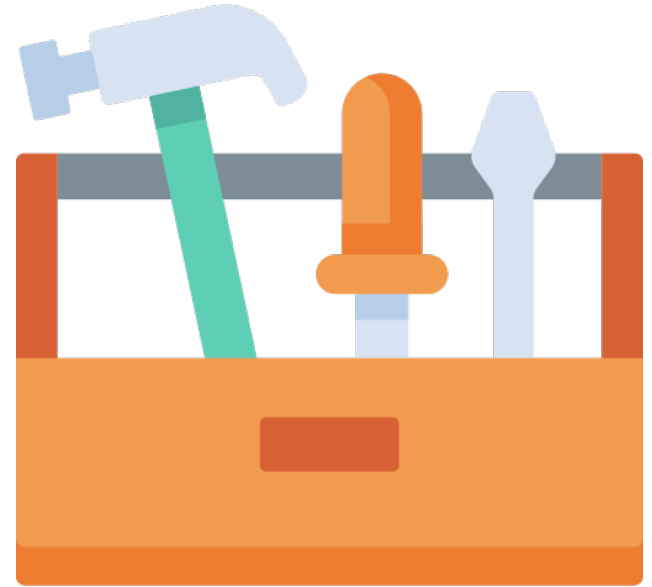
```
def count_by_category(items):  
    pass
```

Input: items = [("fruits", "apple"), ("vegetables", "carrot"),
("fruits", "banana")]

Output: {"fruits": 2, "vegetables": 1}

Balancing AI with Independent Learning

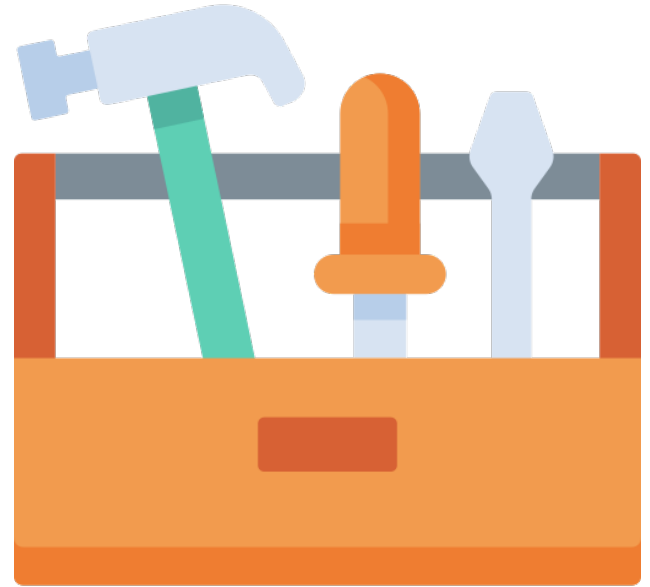
Use AI as a supplement, not a replacement, for your own problem-solving!



Balancing AI with Independent Learning

Use AI as a supplement, not a replacement, for your own problem-solving!

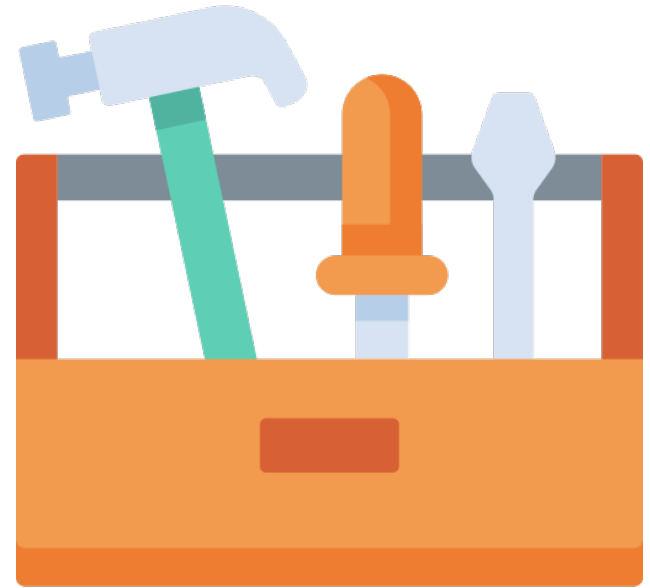
- * **Prompt engineering:**
Giving explicit instructions with each prompt to assist you better.



Balancing AI with Independent Learning

Use AI as a supplement, not a replacement, for your own problem-solving!

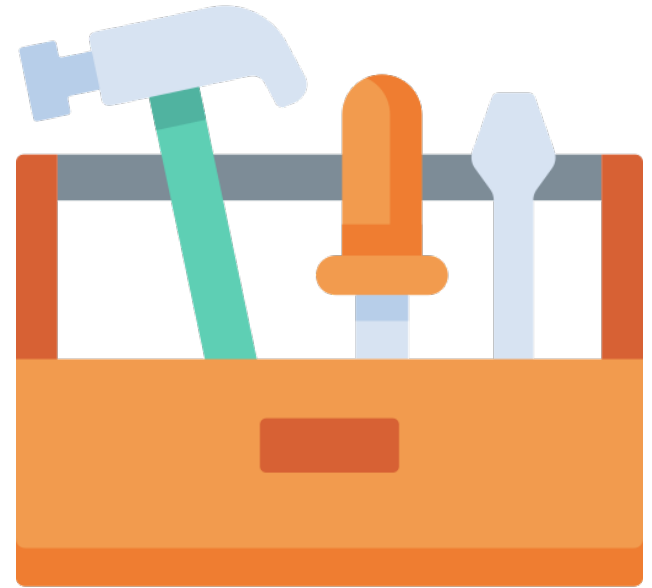
* **Role engineering:** "Please act as a TA, and guide me through this question without giving the solution right away."



Balancing AI with Independent Learning

Use AI as a supplement, not a replacement, for your own problem-solving!

- * **Debugging:** Sharing your code with AI for debugging can speed up your efficiency without taking away from your learnings.

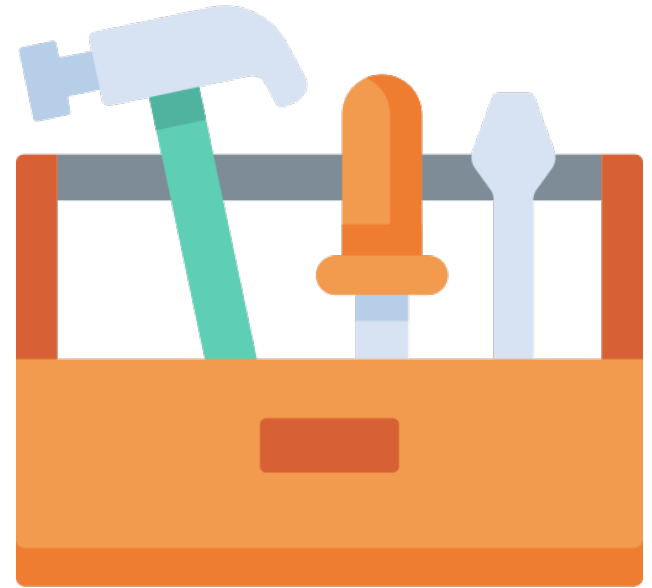


Balancing AI with Independent Learning

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* **Optimizing Solutions:**

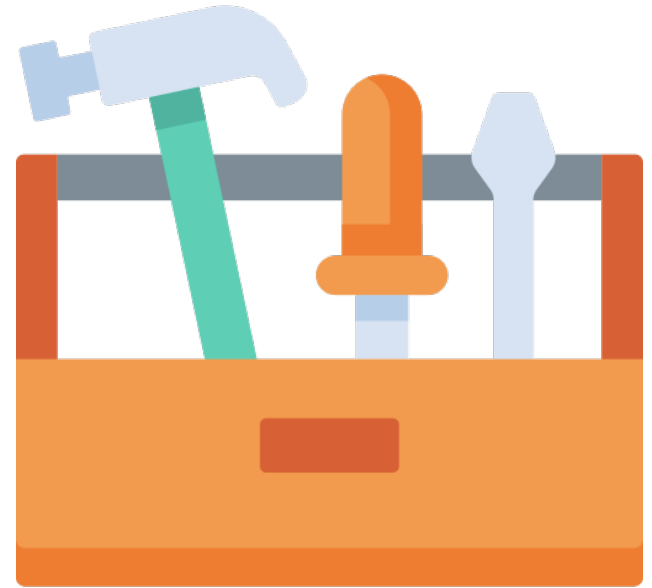
Even if you come up with a working solution on your own, AI can be helpful in improving your solution's efficiency.



Balancing AI with Independent Learning

Use AI as a supplement, not a replacement, for your own problem-solving!

- * **Back and forth:** If you are absolutely hitting a wall in your studies, try spending 20-30min between allowing yourself to use AI.



Break Time!

Take 5 mins to step away from the computer. Feel free to turn off your camera and return promptly after 5 mins.



60 minutes

Collaborative Problem Solving

Choosing Your Breakout Room

1. We'll share the room assignments.
2. We'll demonstrate how to find your room assignment and navigate to your room.





Breakout Rooms

60 minutes

1. 🏃 Navigate to your breakout room.
2. 🙌 Introduce yourselves!
3. 🧑💻 Work on the problems for your selected unit.

10 minutes

Wrap Up



One Good Thing

Let's reflect on today's session. Maybe you got a question answered, overcame a specific coding challenge, or learned how to do something completely new!

In the chat, share at least one good thing you've learned or experienced during the session today.

Let's celebrate together!



Session Survey Time!





Final Reminders

- ☐ Finish the problem sets for additional practice
- ☐ Take the weekly HackerRank assessment by Sunday @ midnight PST
- ☐ Visit a **Study Hall Session** to deepen your understanding, work through challenges, and connect with course material alongside your peers!